Question Paper Discussion

27 September 2025 09:42

LA MARTINIERE FOR BOYS
HALF YEARLY EXAMINATION 2025-26

SUBJECT: COMPUTER APPLICATIONS **TIME:** 2 hrs + Reading Time: 15 mins
CLASS: IX **FULL MARKS:** 100

SECTION A
(Answer All Questions)

Question 1. Select the correct option from the following questions. **(1x20=20)**

i) What is the correct way to declare an integer variable named age and initialize it with the value 15?
a) int age = 15;
b) age = 15;
c) integer age = 15;
d) declare int age = 15;

ii) What is the output of the following code snippet?

```
int x = 5;
System.out.println(x++);
```

a) 4
b) 5
c) 6
d) Error

iii) Which of the following is not an OOPS concept in Java?
a) Polymorphism
b) Inheritance
c) Compilation
d) Encapsulation

iv) Which of the following is a valid Java variable name?
a) 1value
b) @total
c) value1
d) class

9- Computer Applications.- 1/7 (LMB)

v) Which one out of these is not a relational operator?

a) >
b) <
c) =
d) ==

vi) Which of the following is not a primitive data type in Java?

a) int
b) String
c) Boolean
d) double

vii) What will be the output of the following code?

```
int x = 2;
switch (x) {
 case 1: System.out.print("One");
          break;
case 2: System.out.print("Two");
         break;
default: System.out.print("Other");
 }
```

a) One
b) Two
c) Other
d) Error

viii) What will be the output?

```
int i;
for (i = 0; i< 3; i++);
System.out.print(i);
```

a) 012
b) 0
c) 3
d) Infinite loop

ix) What will be the output?

```
for (int i = 1; i<= 5; i++) {
if (i == 3)
break;
```

```
System.out.print(i + " ");
}
```

a) 1 2 3 4 5
b) 1 2
c) 3 4 5
d) 1 2 3

x) Which statement can be used to skip the current iteration in a for loop?
a) skip
b) break
c) continue
d) pass

xi) The syntax of a ternary operator is:

a) (condition) ? (expression)
b) condition ? expression1 : expression2
c) if (condition) then expression1 else expression2
d) ?condition :

xii) Which of the following is equivalent to:

```
if (a > b)
max = a;
else
max = b;
```

a) max = a ? a >b : b;
b) max = b ? b >a : a;
c) max = (a > b) ?a : b;
d) max = (a < b) ?a : b;

xiii) Which of the following is NOT true about if-else statements?

a) They can be nested
b) They can be replaced with a ternary operator for simple conditions
c) They must end with break
d) They execute one of two code blocks

xiv) What is the output?

```
int a = 3, b = 6;
System.out.print((a > b) ? "a" : "b");
```

a) a
b) b
c) 6
d) Error

xv) Which of the following conditions is correct for checking if n is not equal to 10?

a) if (n != 10)
b) if (n =! 10)
c) if n != 10
d) if n <> 10

xvi) What is the output of:

```
System.out.println(15 / 4);
```

a) 3.75
b) 3
c) 4
d) Error

xvii) What is the output of:

```
System.out.println(18 / 4 * 2);
```

a) 9
b) 8
c) 6
d) 2

xviii) The code obtained after compilation is known as:

a) source code
b) object code
c) machine code
d) java byte code

xix) The number of bits occupied by int datatype

a) 4bit

xix) The number of bits occupied by int datatype

a) 4bit
b) 8 bits
c) 16 bits
d) 32 bits

xx) The expression which uses && operator is known as:
a) relational
b) logical
c) arithmetic
d) assignment

Question 2. Answer the following questions [10x2=20]

i. What is the difference between a break statement and a continue statement when they occur in a loop?
skips rest statements and exits loop
skips current iteration

ii. What is the result stored in x, after evaluating the following expression
int x=5; x=++x * 2 + 3 * x--; 30

iii) Analyse the following program segment and determine how many times the loop will be executed and what will be the output of the program segment.

3 times → 1 2 3

```
int i;
for( i=1;i<=3 ; ++i)
System.out.println(i);
```

iv) Give an example of a syntax error. if (i==3),

v) What is the purpose of default in a switch? none of the cases are executed

vi) What will be the output of the following code?

```
float x = -5.27;
System.out.println(Math.ceil(x));   -5
System.out.println(Math.floor(x));  -6
```

vii) What is an infinite loop? (Write an infinite loop statement.) executes infinitely as no condn is set ? for(i=1;;i++)

viii) Differentiate between = and == operators.
assignment operator used to assign value to variable
relational operator used to compare 2 variables for equality

ix) Write a statement in Java for $E = \sqrt{dr^3}$
E = Math.sqrt(d * Math.pow(r,3));

x) Find the output :

```
for(i=1;i<=3;i++)
System.out.print(i);
System.out.print(i);
```

1233

SECTION B
(ANSWER ANY 4 QUESTIONS)

(4x15=60)

Q3) Using the switch write a menu driven program to:

i) To accept a number and print Buzz number or not.
[A "Buzz number" refers to a number that satisfies one of two conditions: It ends with the digit 7 or is divisible by 7]

A number is considered a Buzz number if either of these conditions is true.

Example:

Sample input: n = 17
Sample output: Number is Buzz

ii) To accept a number and print Abundant number or not.
[An abundant number is a positive integer where the sum of its proper divisors (divisors excluding the number itself) is greater than the number itself.]

For example, 12 is abundant because the sum of its proper divisors (1 + 2 + 3 + 4 + 6 = 16) is greater than 12.

Example:

Sample input: n =12
Sample output: Number is Abundant

For an incorrect choice an appropriate error message should be displayed.

Q4) Write a menu-driven program in Java using a **switch statement** to perform the following operations on a series of data:

Menu:

1. Generate the sum of the series
   S=1+3+5+7+... (n terms)

2. Generate the series
   1.5,3.0,4.5,6.0,.....,12.

The program should accept the user's choice and perform the required operation.

Q5) Write a program to input the distance travelled in (kilometers) by a taxi, then calculate and display the total fare depending on the slab given below:

For first 5 km → Rs. 120 (fixed)
For next 10 km → Rs. 15/km
For next 15 km → Rs. 12/km
For distance above 30 km → Rs. 10/km

Q6) Write a program in Java to accept **n** integers from the user. The program should then:

Count and display the total number of **positive** numbers entered.
Count and display the total number of **negative** numbers entered.

Calculate and display the **sum of all positive** numbers entered.

Q7) A courier company charges differently for "ORDINARY" Booking and "EXPRESS" Booking based on the weight of the parcel as per the tariff given.

Weight	Ordinary Booking	Express booking
up to 100 gm	Rs.80	Rs.100
101 gm to 500 gm	Rs.150	Rs.200
501 gm to 1000 gm	Rs.210	Rs.250
more than 1000 gm	Rs.260	Rs.300

Write a program to input weight of a parcel and type of booking ('O' for ordinary and 'E 'for Express).
Calculate and print the charges accordingly.

Q8) Write a program to generate and print all two–digit numbers whose digits differ by **1 or –1**.

Examples:

12 →Difference = 1 – 2 = –1
21 → Difference = 2 – 1 = 1
98 → Difference = 9 – 8 = 1

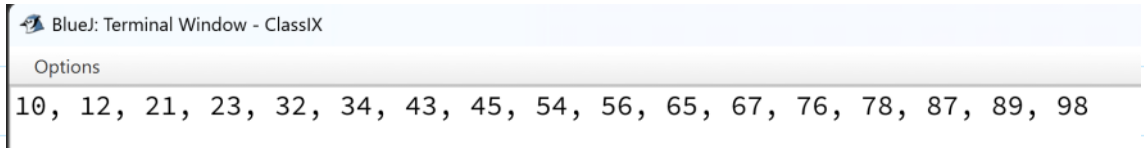
So, the valid numbers will be:
10, 12, 21, 23, 32, 34, 43, 45, 54, 56, 65, 67, 76, 78, 87, 89, 98


```
package Sem1ClassIX;
```

```
package Sem1ClassIX;


/**
 * Write a description of class Q8 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q8
{
    public static void main(String[] args){
        System.out.print("\f");
        int i;
        for(i = 10; i < 98; i++){
            int d1 = i % 10;
            int d2 = i / 10;
            if(d1 - d2 == 1 || d1 - d2 == -1){
                System.out.print(i+", ");
            }
        }
        System.out.print(i);
    }
}
```

Output:




```
package Sem1ClassIX;
import java.util.*;

/**
 * Write a description of class Q7 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q7
{
    public static void main(String args[]){
```

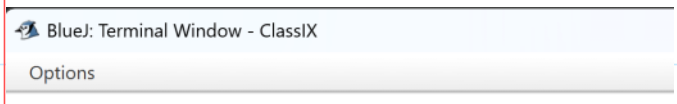
```
public class Q7
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("\f");
        char ch;
        int wt, cost = 0;
        System.out.print("\nPress 'O' for Ordinary");
        System.out.print("\nPress 'E' for Express");
        System.out.print("\n Choose ('O' or 'E'): ");
        ch = sc.next().charAt(0);
        switch(ch){
            case 'O':
                System.out.print("\nEnter the Weight of the parcel: ");
                wt = sc.nextInt();

                if(wt <= 100){
                    cost = 80;
                }
                else if(wt <= 500){
                    cost = 150;
                }
                else if(wt <= 1000){
                    cost = 210;
                }
                else{
                    cost = 260;
                }
                break;
            case 'E':
                System.out.print("\nEnter the Weight of the parcel: ");
                wt = sc.nextInt();

                if(wt <= 100){
                    cost = 100;
                }
                else if(wt <= 500){
                    cost = 200;
                }
                else if(wt <= 1000){
                    cost = 250;
                }
                else{
                    cost = 300;
```

```
                else{
                    cost = 300;
                }
                break;
            default:
                System.out.print("\nWrong Option Selected!");
                System.exit(0);
        }

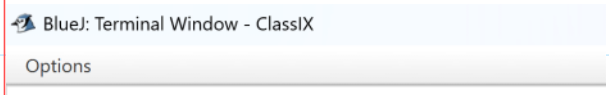
        System.out.print("\nCharges: " + cost);
    }
}
```




```
Press 'O' for Ordinary
Press 'E' for Express
 Choose ('O' or 'E'): o

Enter the Weight of the parcel: 710

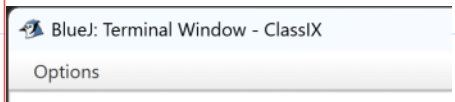
Wrong Option Selected!
```




```
Press 'O' for Ordinary
Press 'E' for Express
 Choose ('O' or 'E'): O

Enter the Weight of the parcel: 600

Charges: 210
```




```
Press 'O' for Ordinary
Press 'E' for Express
 Choose ('O' or 'E'): o

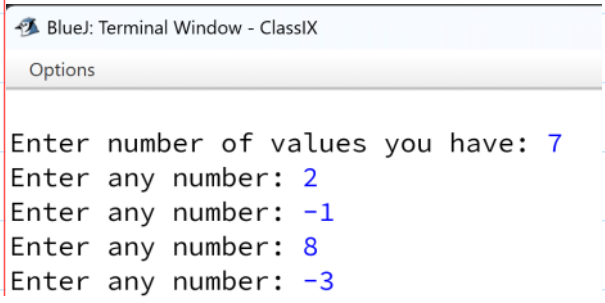
Wrong Option Selected!
```

```
package Sem1ClassIX;
import java.util.*;
```

```
package Sem1ClassIX;
import java.util.*;


/**
 * Write a description of class Q6 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q6
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("\f");
        int num, n, i, sum = 0, cp = 0, cn = 0;
        System.out.print("\nEnter number of values you have: ");
        n = sc.nextInt();
        for(i = 1; i <= n; i++){
            System.out.print("Enter any number: ");
            num = sc.nextInt();
            if(num >= 0){
                cp++;
                sum += num;
            }
            else{
                cn++;
            }
        }
        System.out.print("\nNumber of positive values: " + cp);
        System.out.print("\nSum of positve values: " + sum);
        System.out.print("\nNumber of negative values: " + cn);
    }
}
```

Output:



```
Enter any number: -1
Enter any number: 8
Enter any number: -3
Enter any number: -4
Enter any number: -7
Enter any number: 3

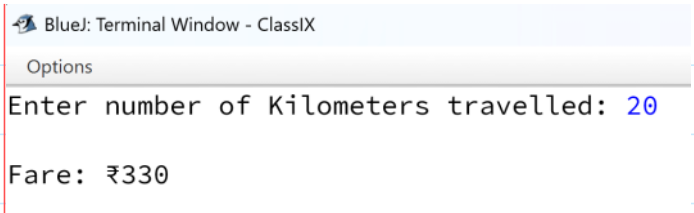
Number of positive values: 3
Sum of positve values: 13
Number of negative values: 4
```

```
package Sem1ClassIX;
import java.util.*;


/**
 * Write a description of class Q5 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q5
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int kms, fare = 120;
        System.out.print("\fEnter number of Kilometers travelled: ");
        kms = sc.nextInt();
        if(kms <= 15){
            fare += (kms-5)*15;
        }
        else if( kms <= 30){
            fare += 150 + (kms - 15)*12;
        }
        else{
            fare += 330 + (kms - 30) * 10;
        }
        System.out.print("\nFare: \u20b9" + fare);
    }
}
```

Output:



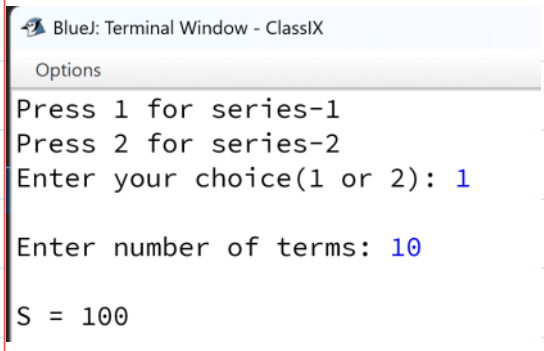


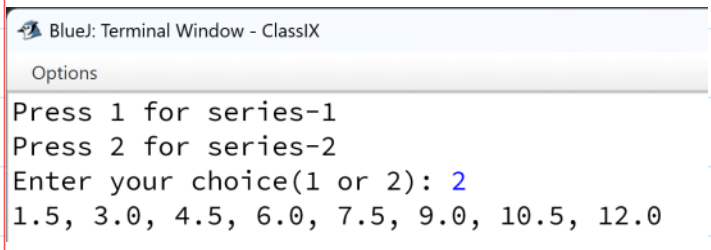

```
package Sem1ClassIX;
import java.util.*;

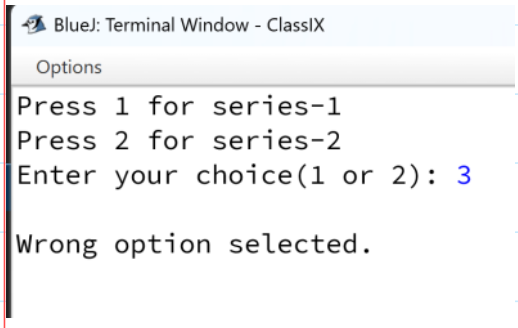

/**
 * Write a description of class Q4 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q4
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("\fPress 1 for series-1");
        System.out.print("\nPress 2 for series-2");
        System.out.print("\nEnter your choice(1 or 2): ");
        int ch = sc.nextInt();
        int n;
        switch(ch){
            case 1:
                System.out.print("\nEnter number of terms: ");
                n = sc.nextInt();
                int sum = 0;
                for(int i = 1; i <= n; i++){
                    sum += (i*2-1);
                }
                System.out.print("\nS = " + sum);
                break;
            case 2:
                double i = 1.5;
                for(; i < 12.0; i+=1.5){
                    System.out.print(i+", ");
                }
                System.out.print(i);
                break;
            default:
                System.out.print("\nWrong option selected.");
        }
```

```
            default:
                System.out.print("\nWrong option selected.");
        }
    }
}
```

Output:








```
package Sem1ClassIX;
import java.util.*;

/**
 * Write a description of class Q3 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q3
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
```

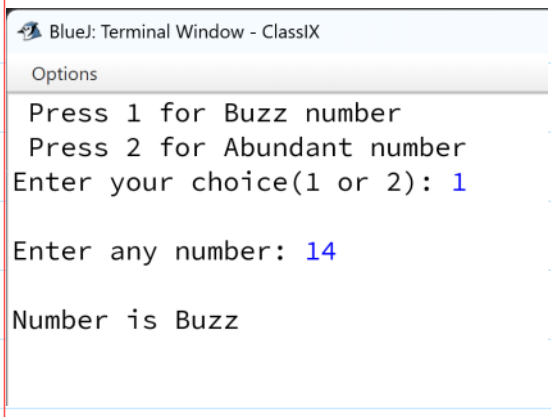
```
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("\f Press 1 for Buzz number");
        System.out.print("\n Press 2 for Abundant number");
        System.out.print("\nEnter your choice(1 or 2): ");
        int ch = sc.nextInt();
        switch(ch){
            case 1:
                System.out.print("\nEnter any number: ");
                int n = sc.nextInt();
                if(n % 10 == 7 || n % 7 == 0){
                    System.out.print("\nNumber is Buzz");
                }
                else{
                    System.out.print("\nNumber is not Buzz");
                }
                break;
            case 2:
                System.out.print("\nEnter any number: ");
                n = sc.nextInt();
                int d, sum = 0;
                for(d = 1; d <= n/2; d++){
                    if(n % d == 0){
                        sum += d;
                    }
                }
                if (sum > n){
                    System.out.print("\nAbundant Number");
                }
                else{
                    System.out.print("\nNot an Abundant Number");
                }
                break;
            default:
                System.out.print("\nWrong Option Selected!");
        }
    }
}
```

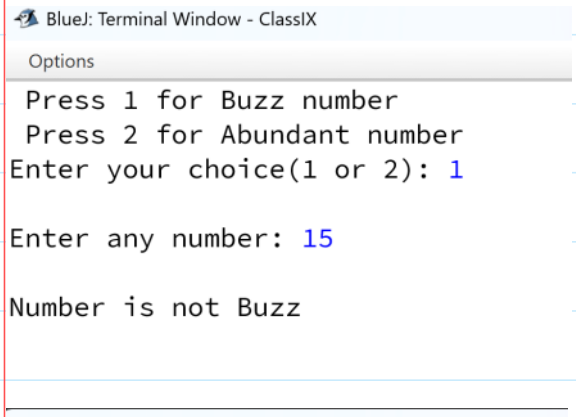
```
BlueJ: Terminal Window - ClassIX
Options
 Press 1 for Buzz number
 Press 2 for Abundant number
Enter your choice(1 or 2): 1
```

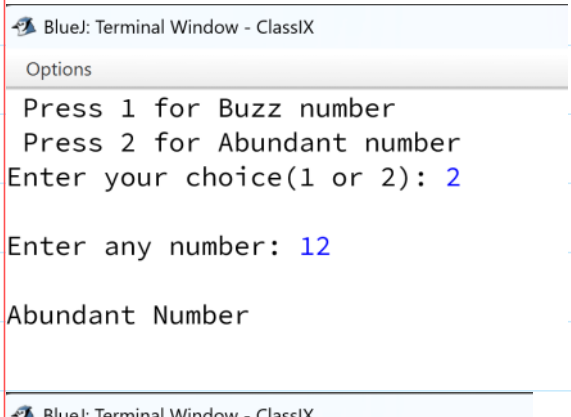
```
 Press 2 for Abundant number
Enter your choice(1 or 2): 1

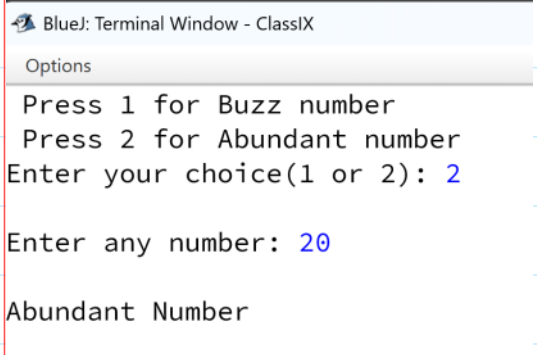
Enter any number: 17

Number is Buzz
```









Abundant Number

